

Project pitch

Team Group 8

Team

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Overview

Our project features a customized version of a Tamagotchi game. The user is able to choose from several animals. The core characteristics of the game involves being able to play, feed, clean and set the pet to sleep. Doing these actions increase their respected vitals. Vitals of energy, mood, hunger and cleanliness deplete over time. The pet dies when two vitals are empty.

- Customizations/extensions: Memory game, guess the number game, animals and snacks.
- Stakeholders:
 - Children (ages 5+ because stakeholder requires basic knowledge of pet care)
 - Future pet owners (allows stakeholder to recognize the responsibility of owning a pet to some degree)
 - Teenagers (the our Tamagotchi project is a game that has a GUI which is visually stimulating)

Functional features

ID	Short name	Description	Champion
F1	Create a new Animal	Program must be able to customize name, colour and type of animal because this is a basic feature for the Tamagotchi game. Program should include animal type of cat, dog and hamster, but must include at least three option types. This is because the options itself are not so relevant, but by giving at least 3 animal options to the user it is necessary to include extensible implementation.	Alejandro
F2	Keep track of vitals	The vital levels must be displayed and updated so that the user is able to decide which behavior to proceed with. The vitals should automatically decrease with time, the rate is relative to the type of vital and the type of animal. This is a creative choice thus not necessary however would differentiate the animals in the program. The following are the vitals that must be included: - Energy - Mood - Hunger - Cleanliness This is because we will have 4 behaviors thus they must correspond to a vital.	Jakub

Functional Features

ID	Short name	Description	Champion
F3	Sleep	The program must include a sleep button, improving his energy on the vitals when pressed instantly. Then the program could disable other buttons but turn the sleep button into wake up button and then can perform any behavior when pressed. Sleep is one of our core behaviors. Vital updates are necessary for user to know, the animal being idle indicates the animal is performing a behavior.	Salvatore
F4	Play	The program must include a play button, he/she is able to play a minigame when pressed. Minigame selection should include two mini games (F8, F9). Once the game is completed the energy vital is decreased, and the mood vital is increased if the user wins or does not change if the player loses. Play is one of our core behaviors. One minigame is a basic requirement, however attempting for two allows helps design system in a way that having multiple minigames is low impact.	Daniele
F5	Feed	The program must include a feed button, he/she can select a snack by clicking on the button with respective snack, improving the pets hunger on the vitals relatively to the snack chosen. The selection should include at least seeds, kibble and fish. This gives each animal at least one food that is best for them. The program must update the vitals instantly and the animal won't be idle while eating. The program won't have in game currency.	Alejandro

Functional requirements

ID	Short name	Description	Champion
F6	Clean	The program must include a clean button and improves the pets cleanliness vital when pressed instantly and the animal won't be idle. A brush could appear on the pet for a couple of seconds. The clean button is one of the core behaviors in our application, the brush is not essential to the system but it would be nice to have for the user.	Daniele
F7	Animal dies upon low vitals	The program must show a death screen that only shows the dead animal with no interaction when the animal dies. This occurs when two vitals are empty. This is important because no actions should be performed when the animal is dead, only to quit the program.	Salvatore

Functional features

ID	Short name	Description	Champion
F8	Memory Game (minigame)	The program must include a memory game, where he/she has to guess and remember a sequence of numbers and repeat them in order. The will screen temporarily shows a sequence of 5 number different every game, after disappearing, the player is prompted to enter the sequence, the game displays won/lost depending if the user is right or wrong. The game could have maximum 3 attempts before losing. This is one of the games that the user is able to select when user presses play.	Jakub
F9	Guess the number (minigame)	The program must include a guessing number game, he/she is prompted to enter a number between 1-10, the screen displays higher/lower or win relatively to the number entered. The winning number is randomly generated, differing for every game, the number is an integer between 1-10. The game could have maximum 3 attempts before losing. Since the range is small on possible numbers, having a maximum attempts is not necessary. This is one of the games that the user is able to select when user presses play.	Jakub
F10	Play again	The program could include a play again button, which restarts the mini game. This is a quality of life feature thus would be nice to have but is not integral to the program	Salvo

Quality requirements

ID	Short name	Quality attribute	Description
QR1	Vitals work properly	Reliability	The games and tasks must correctly affects the vitals, when two vitals reach the value 0, the pet dies flawlessly. The vitals functionally must increase/decrease relatively to the pet without any unpredictable malfunctions. The vitals must decrease with time without interruptions or unexpected behaviours.
QR2	Simple interactions with pets	Usability	The games should be straightforward and tasks must be simple to perform by every person older than 5. The buttons must adequately demonstrate the operation they perform, the activities are basic, they should not require any form of knowledge or experience in order to be performed.
QR3	Responsive activities and games	Responsiveness	The activities must correctly change the vitals. The games must allow the user to play over and over again, they must show the adequate response to the answer given.
QR4	Extendable features	Maintainability	The game functionalities should be simple to extend by the developers. The developers should easily be able to add new pets and colors. The developers could comfortably implement additional mini games.

Time log

Team number	8		
Member	Activity	Week number	Hours
Salvo	Do some research on the original Tamagotchi game	1	2
Alex	Think of possible mini games to implement	1	2
Jakub	Make a mockup to show a general idea	1	2
Daniele	Describe main type of users - stakeholders	1	2
Salvo	Define functional features	1	1
Alex	Define functional features	1	1
Jakub	Define functional features	1	1
Daniele	Define functional features	1	1
Salvo	Define quality requirements	2	1
Alex	Define quality requirements	2	1
Jakub	Define quality requirements	2	1
Daniele	Define quality requirements	2	1
Salvo	Write an overview	2	1
Alex	Write an overview	2	1
Jakub	Write an overview	2	1
Daniele	Write an overview	2	1
		TOTAL	20

Signed contract

<https://docs.google.com/document/d/1gtV2GfB8AUsl8QFnNVNCj2m6FIPV5-FnjsnDaFLvhDw/edit?usp=sharing>